



GE Medical Systems

Technical Publication

Direction 2344544-100

Revision 0

GE Medical Systems

LightSpeed Ultra/Plus HSDCD TMV Upgrade

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

このサービスマニュアルには英語版しかありません。

GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。

この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

本维修手册仅存有英文本。

非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。

未详细阅读和完全了解本手册之前，不得进行维修。

忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
0	6/03/2002	Initial Release of Document. Satisfies CQAs: • 1020467 • 1021249 • 1022369 • 1023481

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Preface

Publication Conventions

Purpose: Standardize conventions for representing information in a uniform way of communicating information to a reader in a consistent manner. Conventions are used so that the reader can easily recognize the actions or decisions that must be made. There are a number of character and paragraph styles used in this publication to accomplish this task. Please become familiar with them before proceeding forward.

Section 1.0

Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

It's important that you read and understand hazard statements, and not just ignore them.



DANGER SEVERE PERSONAL INJURY

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **ELECTROCUTION**
- **CRUSHING**
- **RADIATION**



WARNING SERIOUS PERSONAL INJURY

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **POTENTIAL FOR SHOCK**
- **EXPOSED WIRES**
- **FAILURE TO TAG AND LOCKOUT SYSTEM POWER COULD ALLOW FOR UN-COMMAND MOTION.**



CAUTION
Minor
Personal
Injury

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- Loss of critical patient data
- Crush or pinch points
- Sharp objects



NOTICE
Property
Damage
Only

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will be destroyed
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

1.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout the Preface chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



LASER
LIGHT

HAZARDS MECHANICAL



HEAT



RADIATION



PINCH



AVOID STATIC ELECTRICITY



PROCEDURAL TAG AND LOCK OUT



WEAR EYE PROTECTION



EYE
PROTECTION

Section 2.0 Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as purpose, example, comment or note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

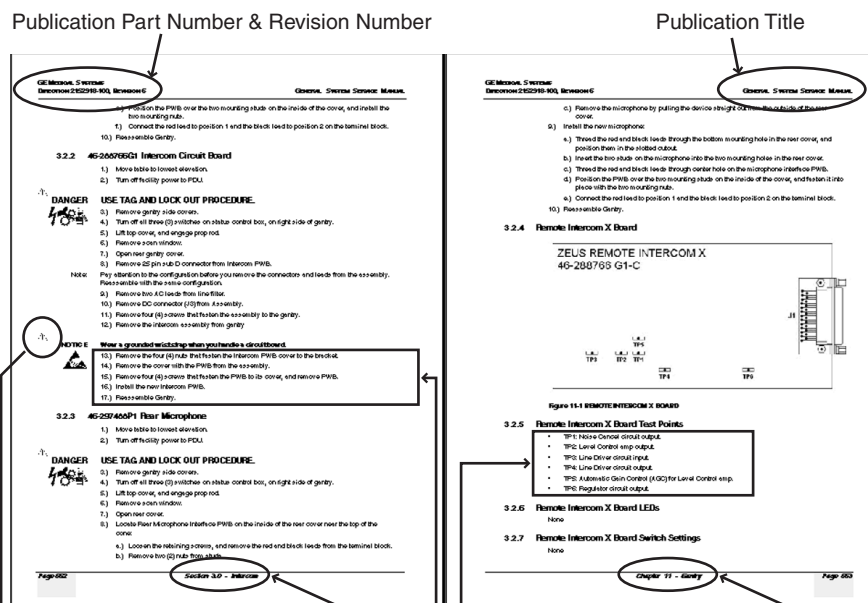
Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, Such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents "additional" information that may or may not be relevant to your situation.

2.2 Page Layout



The current section and its title are always shown in the footer of the left (even) page.

An exclamation point in a triangle is used to indicate important information to the user.

Paragraphs preceded by **Alphanumeric** characters (e.g. numbers) contain information that must be followed in a **specific order**.

The current chapter and its title are always shown in the footer of the right (odd) page.

Paragraphs preceded by a **symbol** (e.g. bullets) contain information that has **no specific order**.

Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

Upgrade Instructions

Section 1.0 Introduction

A study was performed that indicated approximately 7% of all LightSpeed Plus and LightSpeed Ultra systems will experience Taxi Modem Violations (TMV) on their HSDCD communications path with some regularity. About half of these systems will experience chronic problems of this nature.

CT Engineering has found the following root causes:

- 1.) The gantry thermostat switches the gantry fans on and off. The inductive kick of the fan motors creates 500 V spikes on the 120 VAC wave form. The spikes generate noise that is picked up by the HSDCD receiver.
- 2.) When high voltage is asserted across the slip rings, a brief (about 5 cycles) spike of noise is generated between the stationary and rotate side grounds at a frequency of approximately 1.5 to 2.5 MHz - the resonant frequency of the cable. This creates noise on the HSDCD ring.

To address these issues, two system modifications have been created and are detailed in this document:

- 1.) To eliminate noise generated by the gantry fan motors, a snubber circuit (bandpass filter) will be added to the gantry thermostat.
- 2.) To eliminate the ground differential noise, two ferrite common mode current suppressors will be added to the HVDC bus cable harness at the brush block.

These two fixes have been shown to *dramatically* reduce the frequency of TMVs for those systems afflicted with the problem.

Note: **A system not experiencing this condition will not benefit from the application of this upgrade.**

The upgrades detailed in sections [Section 2.0](#) and [Section 3.0](#) may be performed separately; however, to obtain the full benefit of the upgrade, both are required.

[See Table 1](#), for a list of the contents of this kit.

Table 1 Part List for 2345224 - H2/W3 HSDCD TMV REDUCTION KIT

Quantity	Part Number	Description
1	46-170088P10	Capacitive Snubber
2	2287894	Ferrite Core
6	46-208758P10	Cable Ties
1	46-170137P1	RTV 162
1	2344544-100	This Document

Section 2.0

Thermostat Snubber

- 1.) Refer to the *System Service Manual* as needed in the performance of these tasks.
- 2.) Remove the left and right gantry side covers.
- 3.) Turn OFF all three switches (Gantry 120 VAC, Axial Drive Enable and HVDC Enable) on the STC backplane.
- 4.) Remove the gantry thermostat from the left side of the gantry. It is mounted with a bracket and held by two screws on either side.
- 5.) Remove the thermostat cover held on by four screws at each corner of the face cover plate.
- 6.) Install the supplied snubber device as shown in [Figure 1](#).

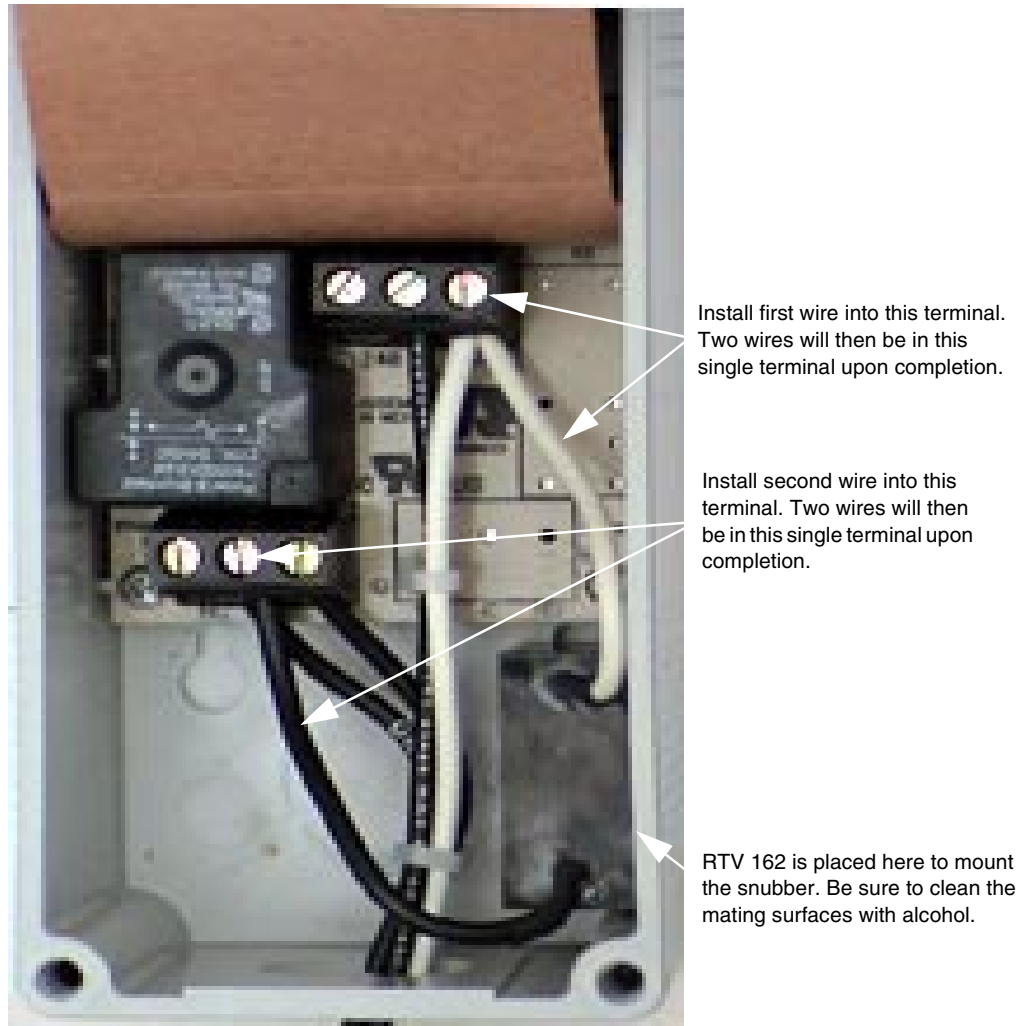


Figure 1 Gantry Thermostat Modification

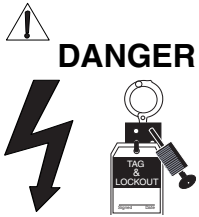
Note: Do not remove any of the existing wires.

- 7.) Clean the snubber housing and the mating surface inside the thermostat housing with alcohol prior to applying RTV 162. Ensure both surfaces are clean and dry.
- 8.) Apply a dab of RTV 162 on the side of the snubber being mounted to the thermostat housing.
- 9.) Stick the snubber to the side of the thermostat housing.

- 10.) Connect the wires as shown in [Figure 1](#).
- 11.) Replace the thermostat cover and remount the thermostat to the gantry.
- 12.) Turn ON gantry 120 VAC at the STC backplane.
- 13.) Reduce the thermostat setting until the fans activate. If fans do not activate, shut off gantry 120 VAC at the STC backplane and verify good connections were made at the terminals inside the thermostat. Begin again at step [11](#).
- 14.) Restore original 26, 1 setting to thermostat.
- 15.) Turn OFF gantry 120 VAC at the STC backplane.
- 16.) Continue to [Section 3.0](#).

Section 3.0

HVDC Cable Harness Ferrite Core Installation



DANGER

USE PROPER LOCKOUT/TAGOUT PROCEDURES.

- 1.) Refer to the *System Service Manual* as needed in the performance of these tasks.
- 2.) Shut down the console software.
- 3.) Turn OFF the power switch at the console.
- 4.) Turn OFF and LOCK-OUT main system power (A1) to the CT Scanner.
- 5.) Remove both gantry side covers.
- 6.) Remove both gantry top covers.
- 7.) Carefully remove the scan window and the gantry rear cover.
- 8.) Remove the upper slip ring cover.
- 9.) Remove the cover plate on the brush block to expose the terminals and lugs of the brush block.
- 10.) Take note that the heavy-gage red wire for the positive DC bus connection is labeled and connected to J1 (nearest to gantry center bore). Notice that the heavy gage black wire for the negative DC bus connection is labeled and connected to J2.
- 11.) Remove the heavy red and black wires at J1 and J2 of the brush block.

Note: **Take care not to drop the lock washes or flat washers from these socketed cap screws into the slip ring during removal.**

- 12.) The crimp lugs for these heavy red and black wires may be too large to fit through the new ferrite cores. If the flared plastic rim of these lugs are too large (plastic edge furthest from the exposed lug), then use a standard pair of pliers to squeeze this plastic rim into an oval shape. It should now be possible to push these red and black wires through the supplied ferrite cores. It may be helpful to push only one lug through the core at a time.

See Figure 2. Install both ferrite cores so that the red and black wires go through both cores. This configuration will provide common mode current rejection.



Slide ferrite cores onto wires as shown. Create a cable tie criss-cross that will hold the first ferrite core in place.

Bind rear most ferrite core to the front most ferrite core with a cable tie.

Figure 2 Ferrite Cores Installed

- 13.) Re-install the socket head cap screws to secure the red wire to J1 and the black wire to J2. Torque these J1 and J2 bolts to 8 N-m (71 inch-lbs, 6 foot-lbs).

Note: **These ferrite cores are insulators so they will not short out the slip ring. To prevent these cores from rubbing against the slip ring and causing wear, follow the next step carefully.**

- 14.) Refer to [Figure 2](#) and [Figure 3](#). The core nearest the brush block needs to be held in place. Loosely install two cable ties in a criss-cross fashion through the ferrite core and around the wires. Pull these ferrite cores away from the slip ring and slide the cores as far up the wires (away from the lugs and brush block) as possible.
- 15.) Tighten the two criss-cross cable ties and cut off the excess. This should secure the ferrite cores up and away from the slip ring.
- 16.) Referring to [Figure 2](#), to help secure the ferrite cores together to eliminate vibration noise, use another tie wrap to loop through both ferrite cores. Again, pull the cable tie tight and remove the excess.
- 17.) [See Figure 3](#). Verify that there is a minimum of 1 cm clearance between the ferrite cores and the slip ring. If needed, pull on the wires a bit to provide this clearance. The wires are stiff enough to hold a new shape. Make sure all wires and cables are clear from the slip ring. Use additional cable ties as needed.

Ensure at least 1 cm clearance between slip ring surface and ferrite cores as shown.

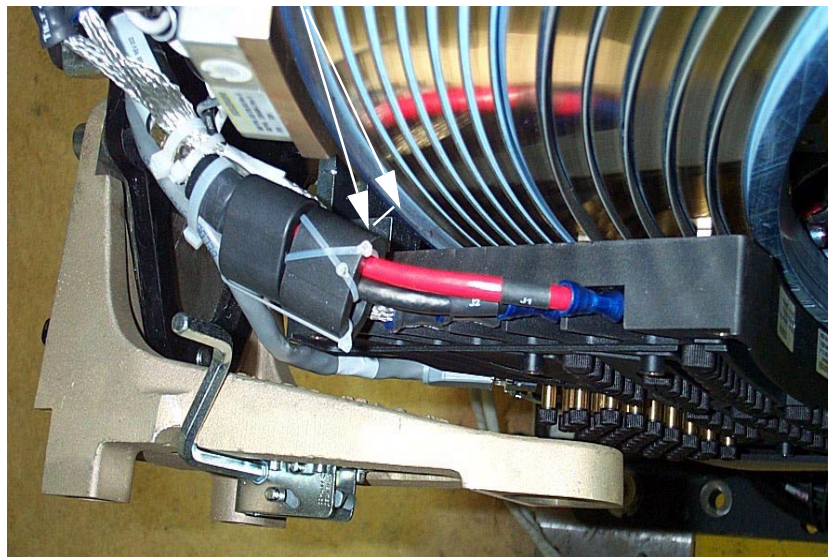


Figure 3 Ferrite Core Clearances

- 18.) Re-install the cover plate for the brush block terminals/lugs.
- 19.) Re-install the slip ring upper cover.
- 20.) Re-install the gantry rear cover and the scan window.
- 21.) Re-install both gantry top covers, turn ON all three switches on the STC backplane, and re-install both gantry side covers.

Section 4.0

Finish Up

- 1.) Restore main system power (A1) to the CT Scanner.
- 2.) Restore power to the console. Allow system to boot to applications.
- 3.) Conduct a test scan to verify CT Scanner operation.



GE MEDICAL SYSTEMS

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